

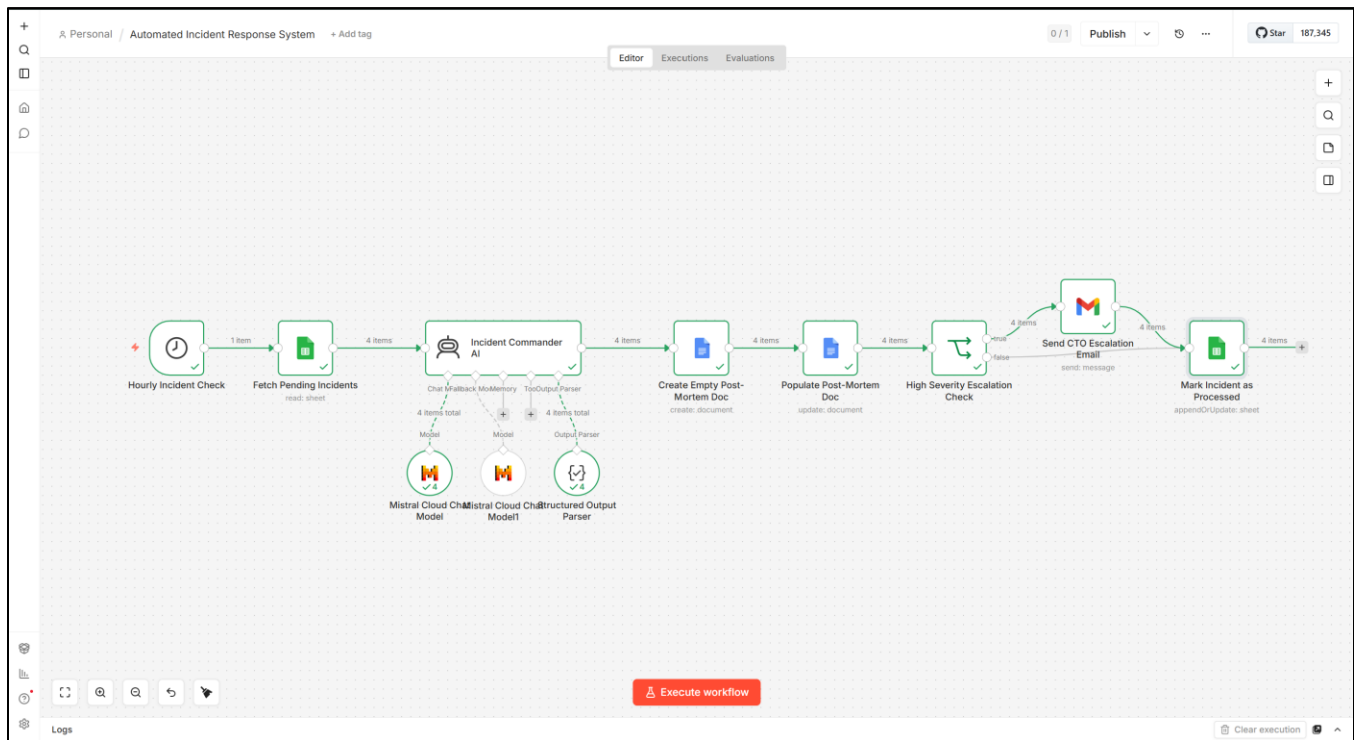
Automated Incident Response & Escalation System

1. Project Overview

This project implements an automated incident response workflow using n8n, Google Sheets, AI, Google Docs, Gmail, and conditional logic. The workflow is designed to run every hour and check a Google Sheet for incidents marked as Pending.

Once a pending incident is found, the AI Agent analyses the raw incident logs, generates a professional post-mortem report, assigns a severity level, and creates an incident title. The workflow then creates a Google Doc report and populates it with the AI-generated incident summary.

The workflow uses an IF node to determine whether escalation is required. If the AI classifies the incident as High, an urgent Gmail notification is sent to the CTO/test email with the incident details and Google Doc link. If the incident is Medium or Low, no email is sent. Finally, all processed incidents are updated in Google Sheets by changing the Report Status to Processed and saving the generated Google Doc link.



Screenshot 1: Full workflow canvas showing all nodes.

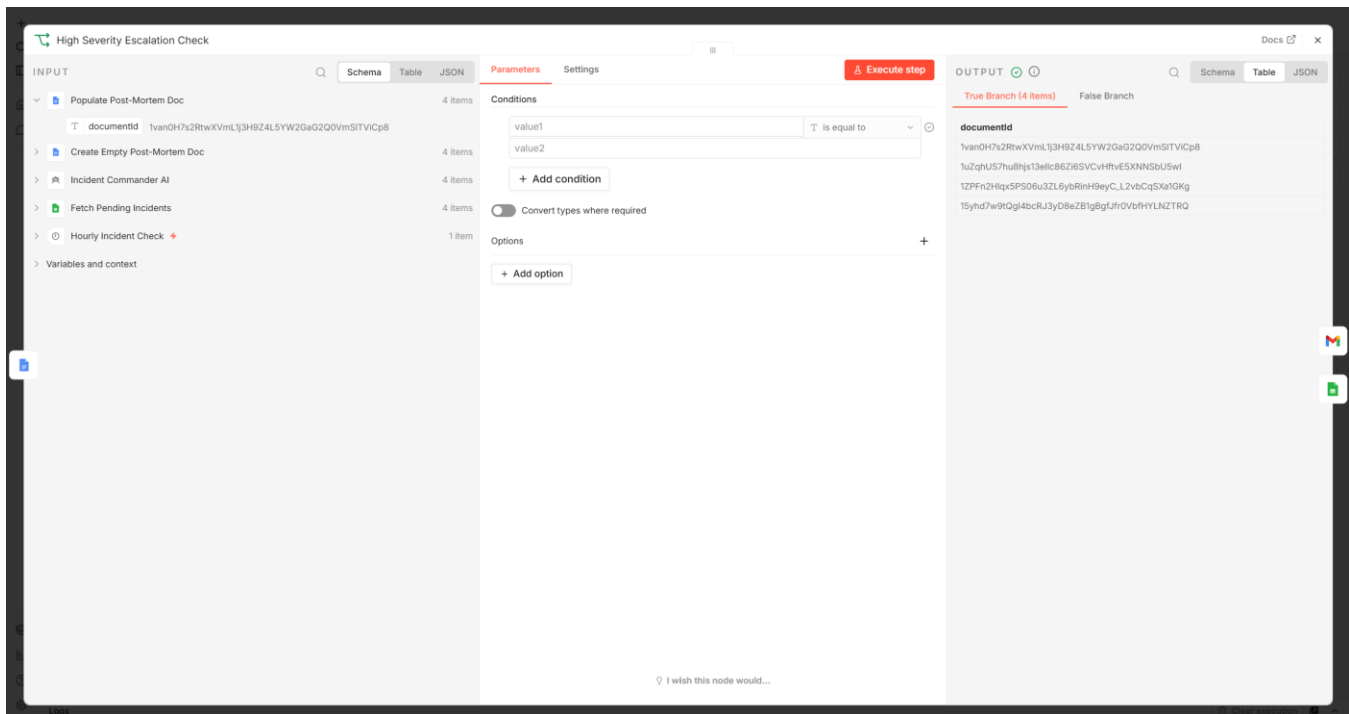
2. System Workflow Operational Flow

The workflow follows the structure below:

Hourly Incident Check

- Fetch Pending Incidents
- Incident Commander AI
- Create Empty Post-Mortem Doc
- Populate Post-Mortem Doc
- High Severity Escalation Check
 - └── TRUE → Send CTO Escalation Email → Mark Incident as Processed
 - └── FALSE → Mark Incident as Processed

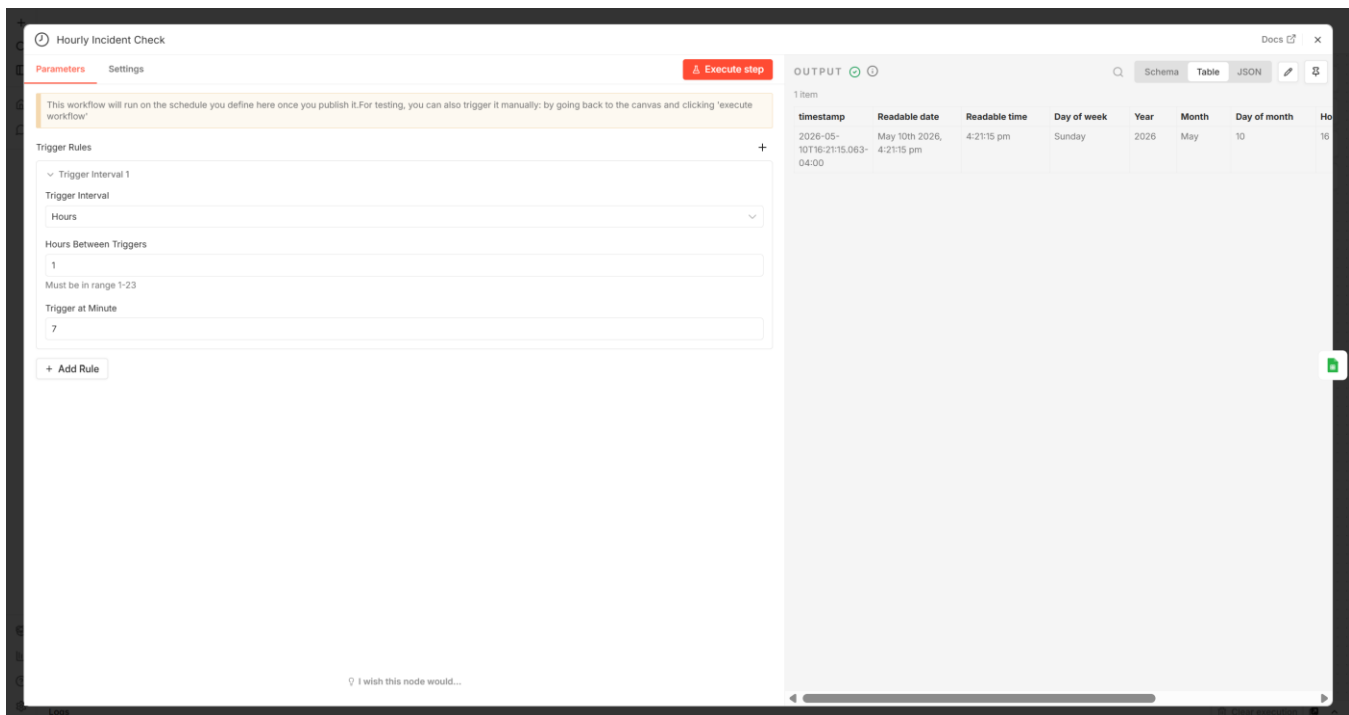
This ensures that every pending incident is documented, only high-severity incidents are escalated, and all incidents are closed out in the Google Sheet after processing.



Screenshot 2: Workflow canvas showing TRUE and FALSE paths from the IF node

3. Schedule Trigger Configuration

The workflow starts with a Schedule Trigger node named Hourly Incident Check. It is configured to run every hour so that new pending incidents are checked automatically.



Screenshot 3: Schedule Trigger configuration

4. AI Agent Configuration

The AI Agent is named **Incident Commander AI**. It analyses the raw incident logs and produces a structured JSON response containing the post-mortem body, severity score, and incident title.

System Prompt

You are an expert Site Reliability Engineer and Incident Commander.

Analyse incident logs and generate a professional post-mortem report.

Classify severity using these rules:

High:

- *Production outage*
- *Customer-facing downtime*
- *Security breach*
- *Data loss*
- *Failed payments or login outage*
- *Major infrastructure failure*

Medium:

- *Degraded performance*

- Partial service disruption
- Repeated warnings that require follow-up

Low:

- Minor alerts
- Self-healing issues
- No customer impact
- Informational warnings

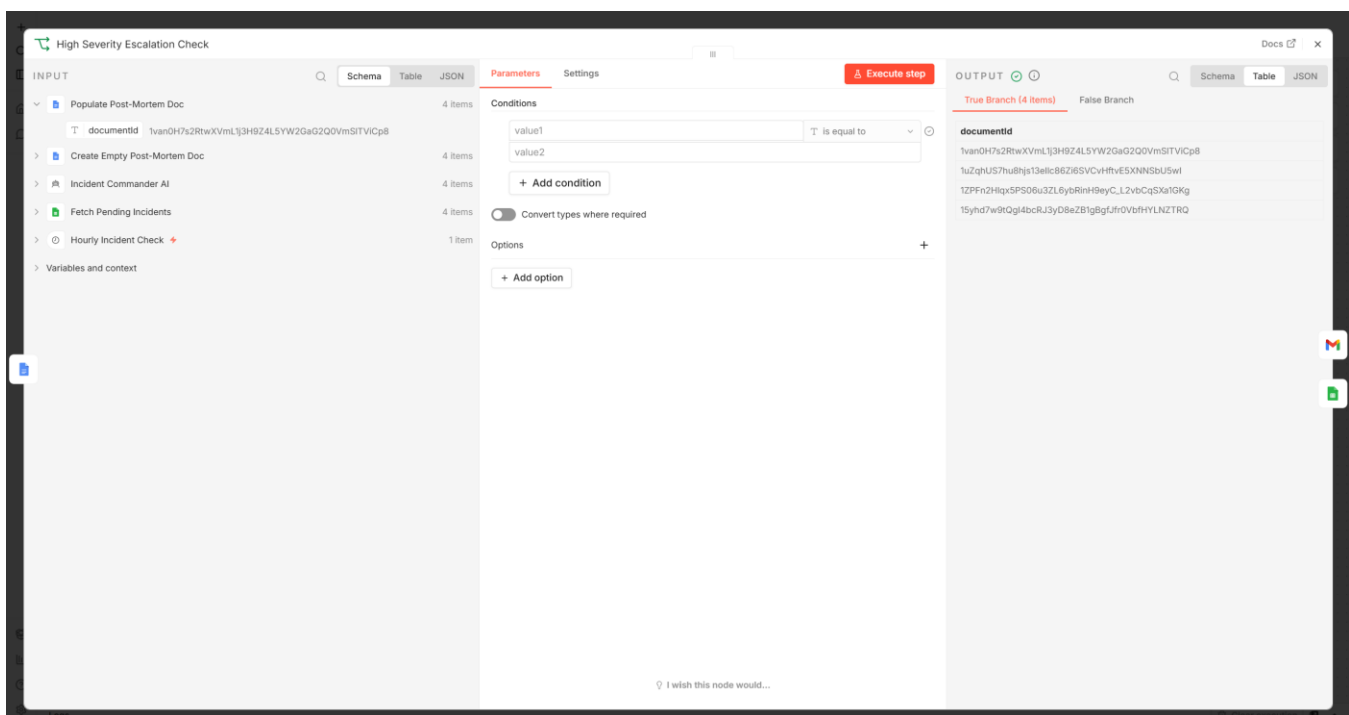
Return valid JSON only.

Do not include markdown, explanations, or extra text..

5. Escalation Logic

The IF node named **High Severity Escalation Check** checks the AI-generated severity_score.

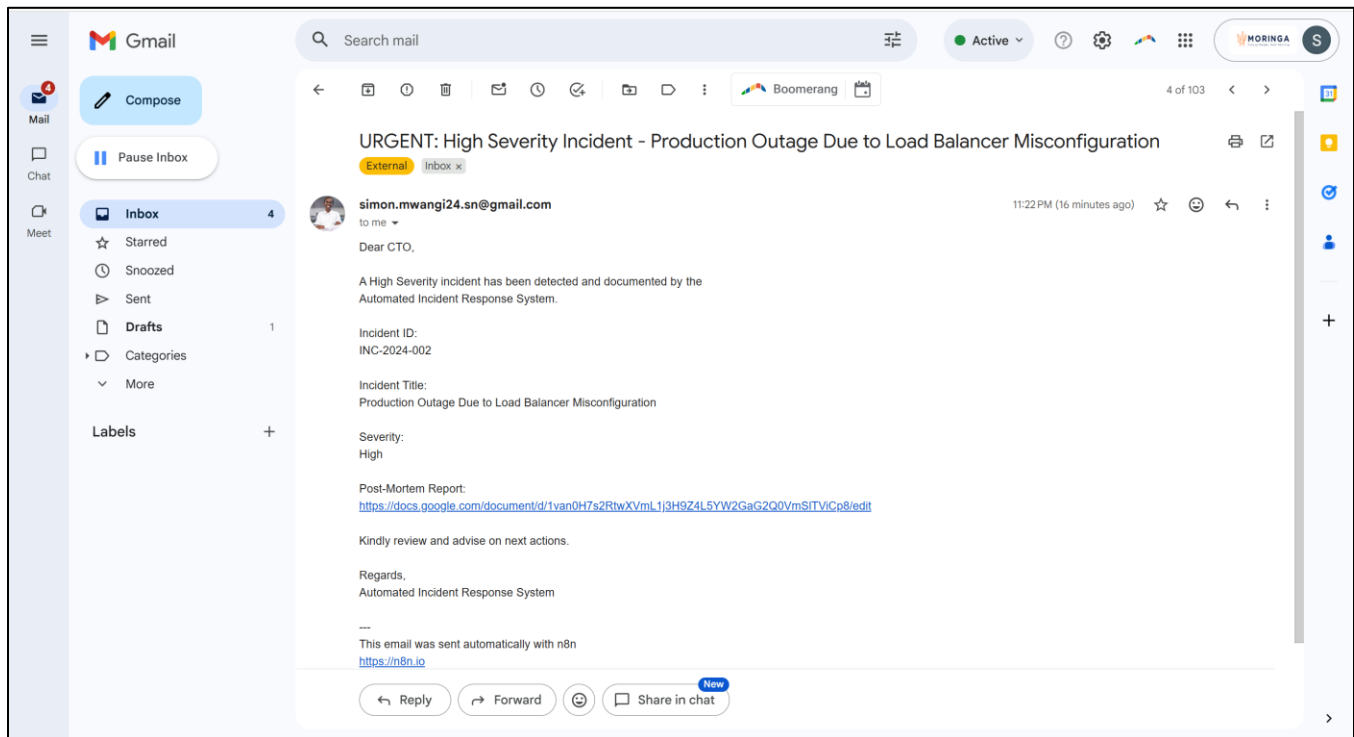
If true, the workflow sends an urgent Gmail escalation. If false, the workflow skips email and proceeds directly to final logging.



Screenshot 4: IF node configuration

6. Gmail Escalation

The Gmail node named **Send CTO Escalation Email** sends an urgent email only when the incident severity is High.



Screenshot 5: Received Gmail escalation email

7. Workflow Final Google Sheets Update

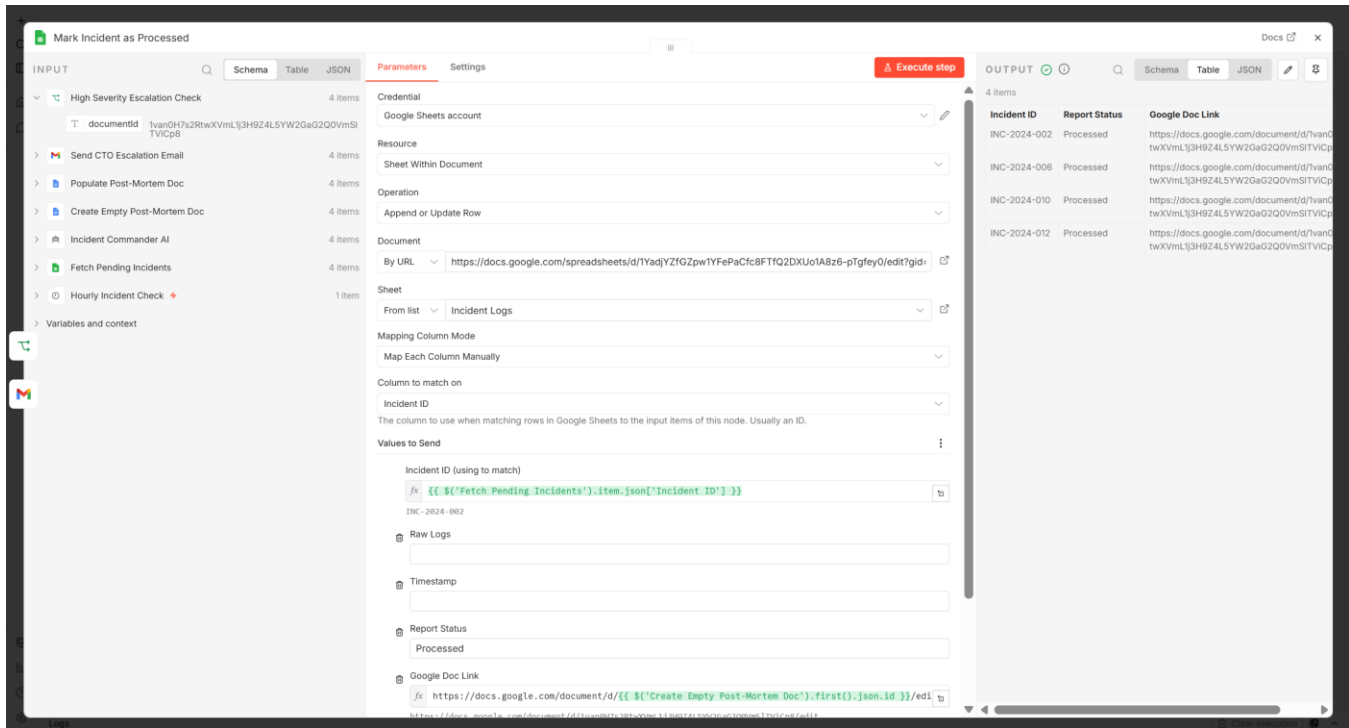
The final Google Sheets node named **Mark Incident as Processed** updates the original incident row.

It sets:

Report Status = Processed

Google Doc Link = Generated Google Doc URL

This closes the loop and prevents the workflow from processing the same incident again.



Screenshot 6: Google Sheets update node configuration

Capstone_Incident_Logs_GoogleSheet - SM

	A	B	C	D	E	F
	Incident ID	Raw Logs	Timestamp	Report Status	Google Doc Link	
1	INC-2024-001	[2024-12-09 02:15:34] WARNING: Disk usage on v	2024-12-09 02:15:34	Processed	https://docs.google.com/document/d/1sc0rlrvqhz-H2T9VlrN9f	
2	INC-2024-002	[2024-12-09 03:42:18] INFO: SSL certificate for de	2024-12-09 03:42:18	Processed	https://docs.google.com/document/d/1van0H7s2RtwXVml1j3	
3	INC-2024-003	[2024-12-09 05:20:45] WARNING: API rate limit n	2024-12-09 05:20:45	Processed	https://docs.google.com/document/d/1IAksLox8RNe4miebQc	
4	INC-2024-004	[2024-12-09 08:15:22] INFO: Scheduled backup fi	2024-12-09 08:15:22	Processed	https://docs.google.com/document/d/1-k2K8ycx4k_tNq12JaFf	
5	INC-2024-005	[2024-12-09 10:30:11] WARNING: Memory usage	2024-12-09 10:30:11	Processed	https://docs.google.com/document/d/1AFyxl_PoLb5KSkmBgL	
6	INC-2024-006	[2024-12-09 01:45:12] ERROR: Payment gateway	2024-12-09 01:45:12	Processed	https://docs.google.com/document/d/1van0H7s2RtwXVml1j3	
7	INC-2024-007	[2024-12-09 04:22:38] ERROR: Database connect	2024-12-09 04:22:38	Processed	https://docs.google.com/document/d/1MMRIQLhm4ssH_ojlr2	
8	INC-2024-008	[2024-12-09 07:10:25] ERROR: Email service (Sen	2024-12-09 07:10:25	Processed	https://docs.google.com/document/d/1Wx7_j6Ca3WZZKYsb2x	
9	INC-2024-009	[2024-12-09 09:35:18] WARNING: Load balancer	2024-12-09 09:35:18	Processed	https://docs.google.com/document/d/1Xmuo70wi9wLDVFZTjV	
10	INC-2024-010	[2024-12-09 11:20:33] ERROR: M-Pesa API integr	2024-12-09 11:20:33	Processed	https://docs.google.com/document/d/1van0H7s2RtwXVml1j3	
11	INC-2024-011	[2024-12-09 00:15:42] CRITICAL: Primary databas	2024-12-09 00:15:42	Processed	https://docs.google.com/document/d/1j7f2rMspE727yEiC3y4V	
12	INC-2024-012	[2024-12-09 06:45:18] CRITICAL: Security breach	2024-12-09 06:45:18	Processed	https://docs.google.com/document/d/1van0H7s2RtwXVml1j3	
13	INC-2024-013	[2024-12-09 12:30:05] CRITICAL: Complete servic	2024-12-09 12:30:05	Processed	https://docs.google.com/document/d/1pRiF6etRai2wfdtTohYr	
14	INC-2024-014	[2024-12-09 14:15:33] CRITICAL: Data corruption	2024-12-09 14:15:33	Processed	https://docs.google.com/document/d/1IXStcT8tOOFWAawJ_Y	
15	INC-2024-015	[2024-12-09 16:40:22] CRITICAL: Payment proces	2024-12-09 16:40:22	Processed	https://docs.google.com/document/d/1-z7GL11SJaDeV0i55mk	
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27						

<https://docs.google.com/spreadsheets/d/1YadY2fGZpw1YFePaCfc8FTfQ2DXUo1A8z6-pTgfeyO/edit?gid=1705521381#gid=1705521381>

Screenshot 7: Google Sheet showing Processed status and Google Doc link

8. Conclusion

The completed workflow successfully automates the incident response process. It reads pending incidents from Google Sheets, uses AI to generate structured post-mortem reports, creates Google Docs documentation, escalates only high-severity incidents by Gmail, and updates the source sheet to mark incidents as processed. This reduces manual reporting effort, prevents repeated processing, and ensures that only critical incidents are escalated immediately.

